Lesson 9

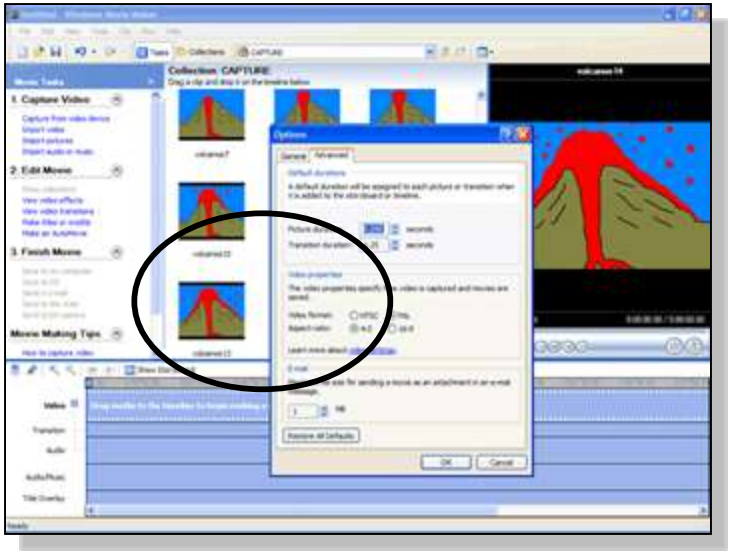
Computer Animation

Early Animation Processes

- *Animation* is created when still images are played in rapid succession so that they appear to produce images that are constantly moving.
- Animation appears to have continuous motion because the human eye (brain) "holds-onto" the still image for just a brief moment after it is viewed, and the image is still "there" (in your brain) when the next image is viewed.
- The timing between individual images must be fast enough for the sequence appear smooth.

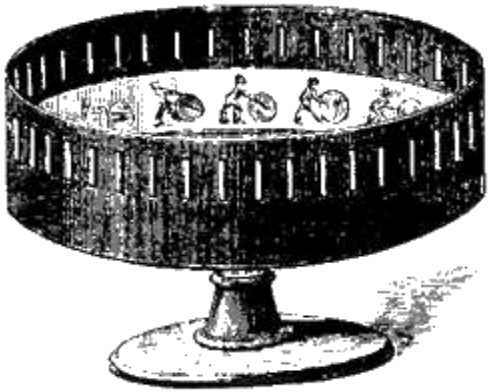
Early Animation Processes

- The National Television Standards Committee (NTSC) frame rate = 30 frames per second for television (North America and Japan).
- PAL (Phase Alternate Line) is the European standard of 25 frames per second.
- The standard rate for film (motion pictures) = 24 frames per second.
- A frame rate of 30 fps will require 1800 images for one minute of animation (30 fps x 60s).



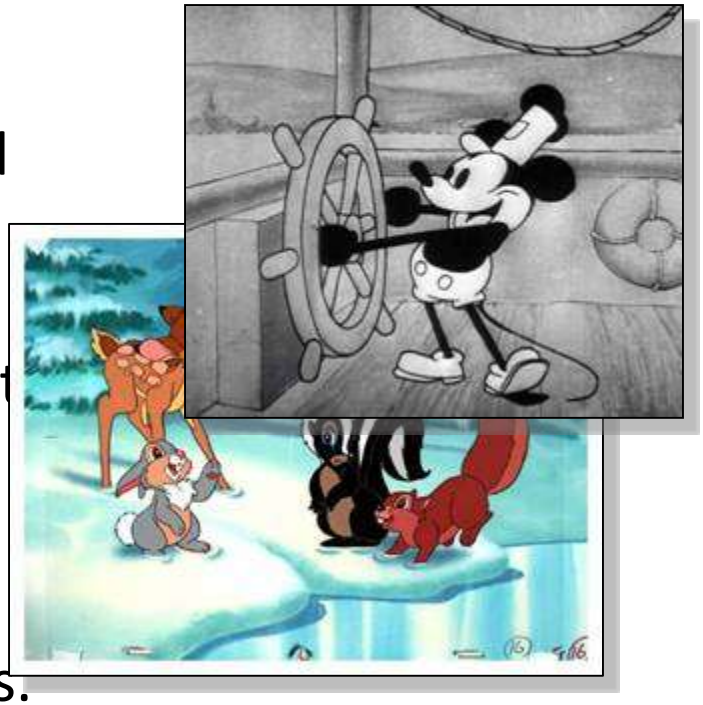
Early Animation Processes

- The *Zoetrope* was a device that was used to produce animation in the 1800s. It consisted of a circular frame holding individual, sequenced images, and a fixed viewpoint through which the spinning pictures were viewed. The term "movies" comes from the moving images.



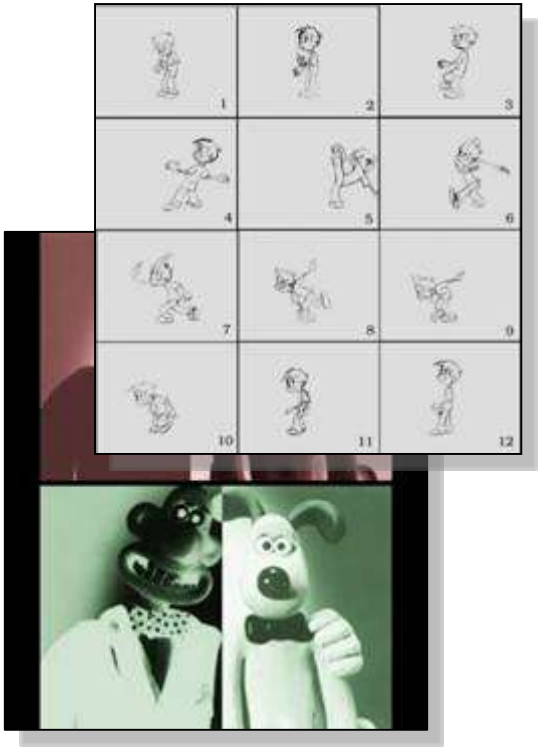
Early Animation Processes

- The Walt Disney Studios developed animation into a modern art during the 1930s and 1940s.
- The different "layers" of the animated scene were painted ont transparent sheets, called *cels*.
- A hierarchy of artists was developed for drawing and painting the sequences of images.



Early Animation Processes

- A master artist would draw the most important or *key* frames (*"keyframes"),* and less-skilled or less-experienced artists would fill in the action for the in-between (*"tweens")* frames. Other artists would paint or fill the outlines with color.
- *Stop-action animation* uses clay or other models whose positions are sequentially altered and photographed for each frame.



Traditional Animation Techniques

- Film animators use a variety of methods for depicting and emphasizing motion sequences.
- These include object deformations, spacing between animation frames, motion anticipation and follow-through, and action focusing.
- During the late 1920's through the 1930's, Walt Disney worked to improve the techniques of his studio animators.
- Disney set up drawing classes for his animators at the Chouinard Art Institute in Los Angeles under Instructor Don Graham.
- Through these lessons and interaction between Disney and his staff, a set of twelve principles was developed.
- These principles were used in Disney animated productions including *Snow White*, *Pinocchio*, *Fantasia*, *Dumbo*, and *Bambi*.

1. Squash and Stretch

- Defines rigidity of material.
- Important in facial animation.
- Living flesh distorts during motion.
- Exaggerated deformations will emphasize motion and impact.
- Although objects deform like rubber, they must maintain volume while being squashed and stretched.
- A bouncing ball will squash or elongate on impact and stretch vertically as it leaves the point of impact.

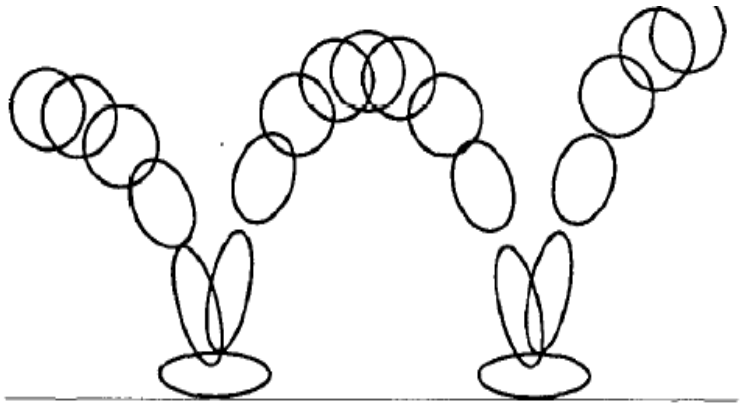
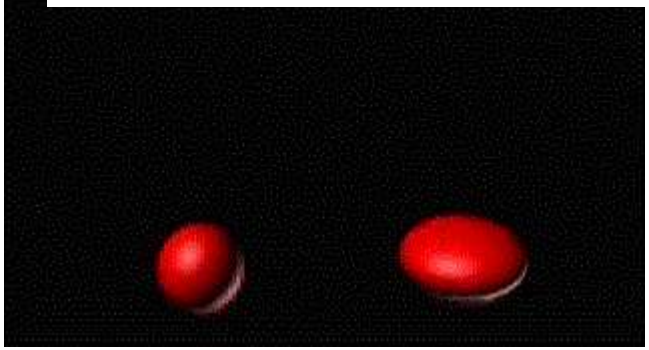


FIGURE 2. Squash & stretch in bouncing ball.



2. Timing and Motion

- Timing is the amount of frames between poses.
- Comedians and actors work with their timing to get the maximum impact from their lines.
- Timing can imply weight. Light objects have much less resistance and usually move much quicker than heavy objects.
- Speed can imply emotion. A fast walk may mean happiness and a slow walk may mean depression.
- Timing gives meaning to movement.
- Proper timing is critical to making ideas readable.

3. Anticipation

Animation can occur before an action. Before you jump, you bend your knees.

By exaggerating this action, the animator can guide the viewer's eyes.

The formula for most animations is anticipation, action, and reaction.

Anticipation is preparation for an action

Example:

Goofy prepares to hit a baseball.

4. Staging

- A clear presentation of an idea.
- The animator can use the camera viewpoint, the framing of the shot, and the position of the characters to create a feeling or strengthen understanding.
- Important that only one idea is seen by the audience at a time.

Some Techniques:

1. Use motion in a still scene or use of static movement in a busy scene.
2. Use of silhouettes (outlines,shadows)

5. Follow Through and Overlapping Action

1. Follow Through

Follow Through is the action that follows the main action. It is the opposite of anticipation

Example: after throwing a ball

2. Overlapping Action

Starting a second action before the first has completed.

Example: Luxo Jr.'s hop with overlapping action on chord.

6. Straight Ahead Action and Pose-to-Pose Action

- Straight Ahead animation means drawing the frames in sequence. This leads to spontaneous motion. It works well with abstract animation.

- Pose To Pose is the more often used animation technique. It requires the animator to create strong posed (keyframes) first and add the in-between frames later.

- Pose to Pose is used for animation that requires good acting, where the poses and timing are all important.

7. Slow in and Out

Spacing of inbetween frames to achieve subtlety of timing and movement.

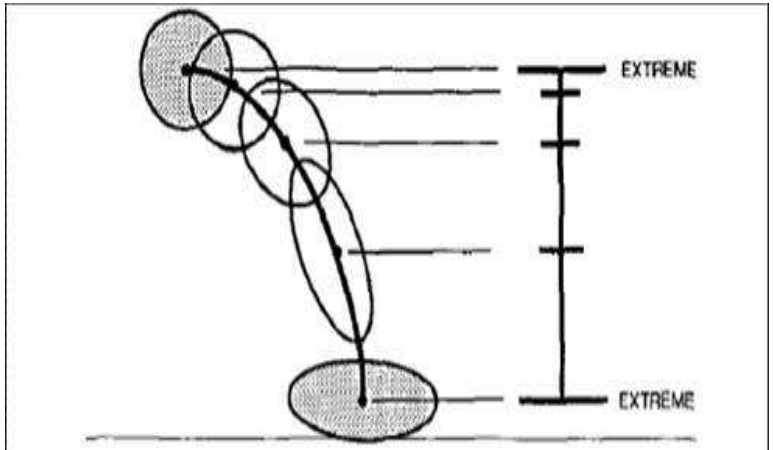
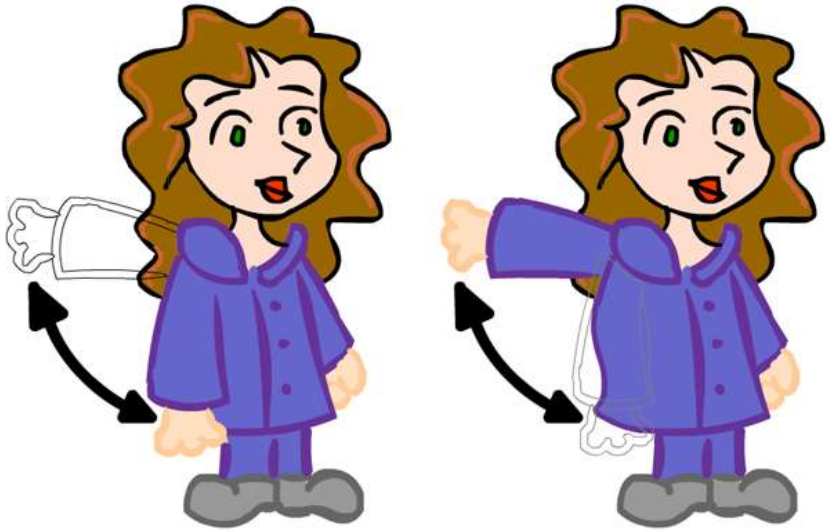


FIGURE 9. Timing chart for ball bounce.

1. Also known as ease in and ease out.
2. Most motion starts slowly, accelerates, and then slows again before stopping.
3. Gravity has an effect on slow in/slow out. When a ball bounces, it increases in speed as it gets closer to the ground. It decreases in speed at the top of the arch

8. Arcs

- Visual path of action for natural movement.
- Makes animation much smoother and less stiff than a straight line.
- In 3D animation, a motion arc is usually created using a spline curve.
- Pivot points often define the arc. The pivot point for the thigh is the hip, and the pivot point for the calf is the knee.



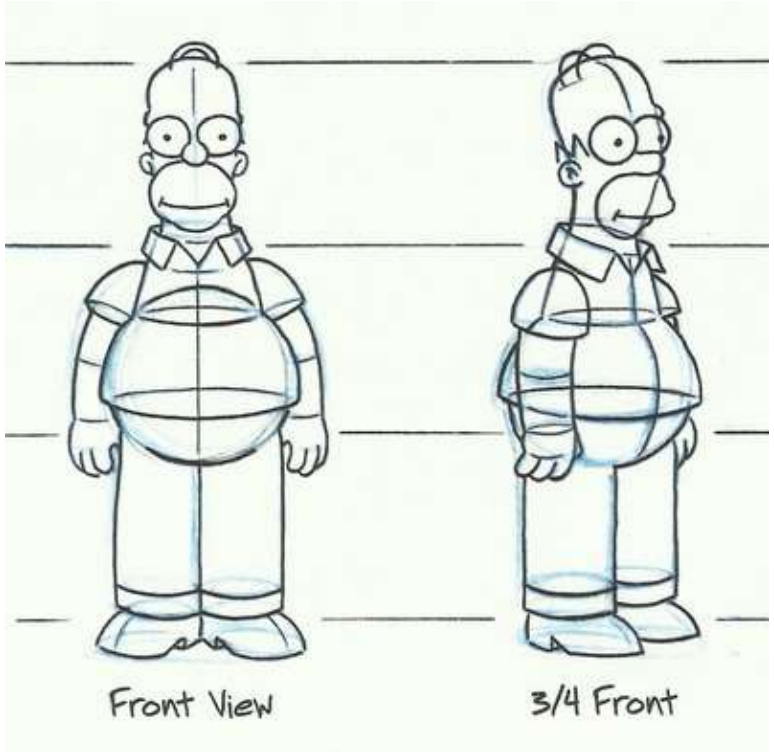
9. Exaggeration

- Accentuating the essence of an idea via the design and the action.
- Exaggeration is used to increase the readability of emotions and actions.
- Individual exaggerated poses may look silly as stills but add dramatic impact when viewed for a split second.
- Animators should be careful to use exaggeration to increase understanding of feeling.
- Be careful not to over-exaggerate everything.

10. Secondary Action

- Action that results directly from another action.
- Used to increase the complexity and interest of a scene.
- **Examples:**
- Most people blink their eyes when they turn their head.
- Facial expressions are secondary actions.
- Secondary actions are also actions caused by the impact of another object. The movement of a ball that has been kicked is a secondary action.

11. Solid Drawing



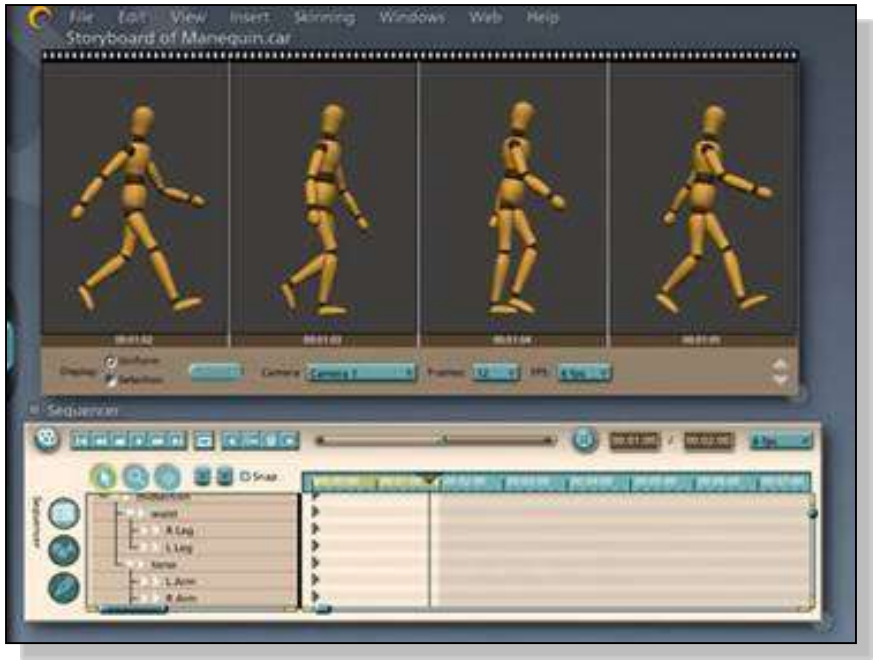
- In 3D animation, this would be equivalent to Solid Modeling.
- To get maximum feeling from the audience, animated characters must be drawn or modeled precisely.
- Proper drawing and modeling can reveal a character's weight, character, and emotion.
- Proper drawing and modeling are needed to give the character proper depth and balance.

12. Appeal

- Refers to what an audience would like to see.
- Character cannot be too simple (boring) or too complex.
- Animated characters need to have a unique personality and have a wide range of emotions (happy, excited, fearful, embarrassed, angry, scared, etc.).
- Character flaws are actually a good thing. Audiences can be sympathetic to characters that have a flaw or two.
- Complex personalities and moral ethical dilemmas add to character appeal.

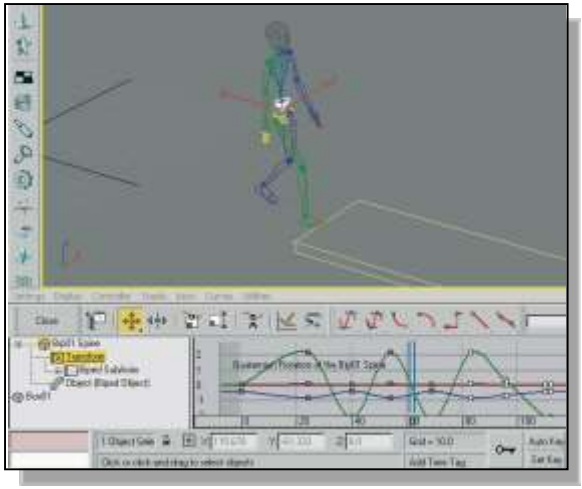
Computer Animation

- Unlike early animation, where every frame must be created to produce movement, in computer animation you define points in time (known as *keyframes*) and the computer draws all of the in-between frames.



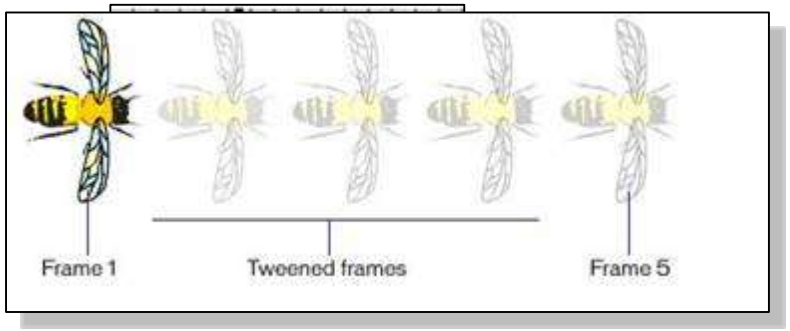
Computer Animation

- Position the object to be animated where you want the action to begin; this is first keyframe.
- Set the number of frames that you want to use for the animation sequence. A one second “movie’ would typically use 30 frames per second (30 fps NTSC); two seconds would use 60 frames, etc.
- Move, scale, or deform the object to become the next keyframe.



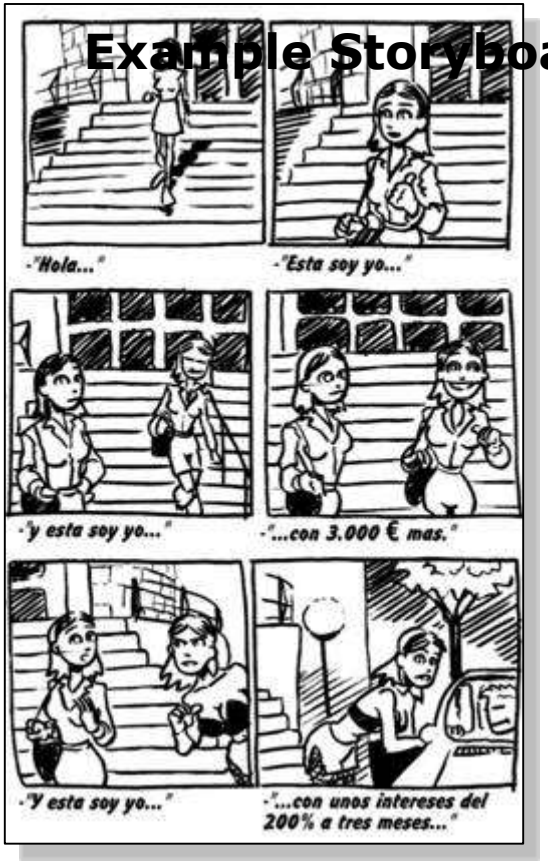
Computer Animation

- Instruct the computer to calculate all of the transformations that will occur between the first keyframe and the last.
- The computer will produce the 28 additional "in-between" images (*tweens*) needed for the one second of animation (you created the other two frames, the keyframes, for a total of 30).



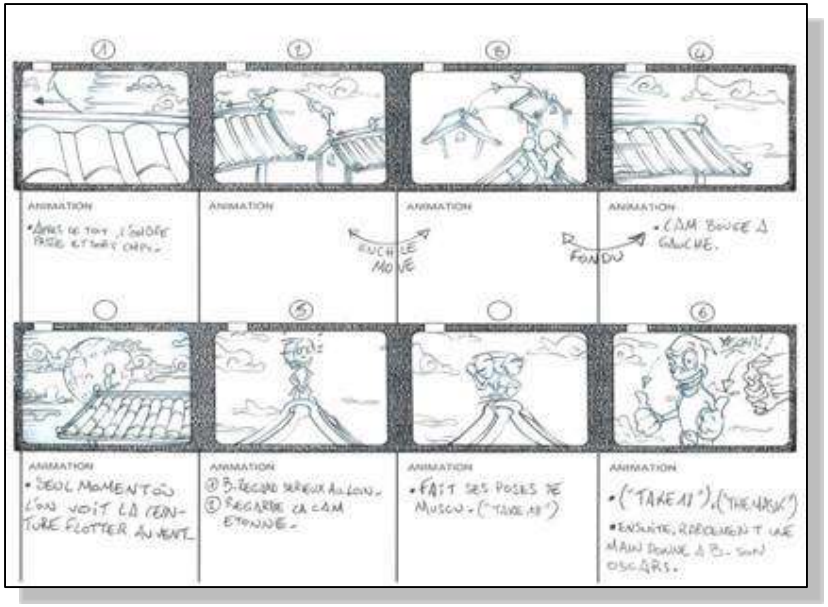
Storyboarding

- A *storyboard* is a graphic, sequential depiction of an animation that is going to be created. It is a visual script designed to make it easier to see the animation scenes before they are created.
- A storyboard identifies the major events in the story and illustrates them in cells (small squares or rectangles), which are drawn out in a sequential pattern.



Storyboarding

- Storyboards are used for movies, TV, commercials, and animation.
- The artwork does not have to be pretty or complex, but it should be neat and comprehensible.
- The pictures in the storyboard should be accompanied by text.



Animation Techniques

- Modeling and animation are loosely coupled.
- Modeling describes control values and their actions.
- Animation describes how to vary the control values.
- There are a number of animation techniques, including the following:
- User driven animation
  - Keyframing
  - Motion capture
- Procedural animation
  - Physical simulation
  - Particle systems
  - Crowd behaviors
- Data-driven animation

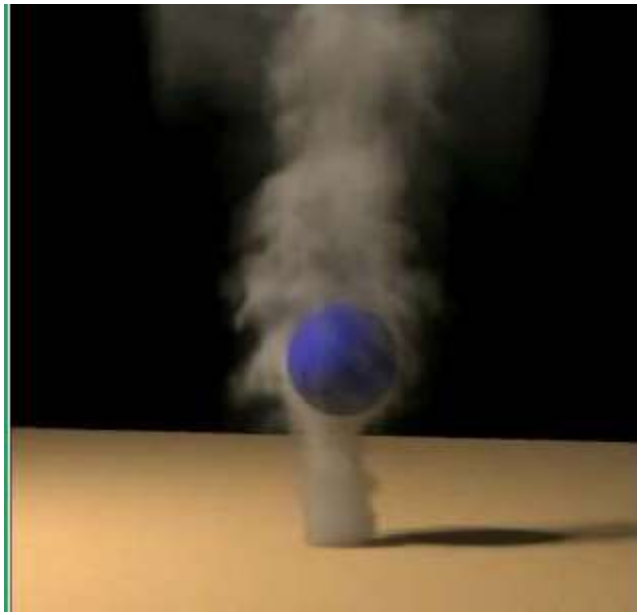
User driven animation

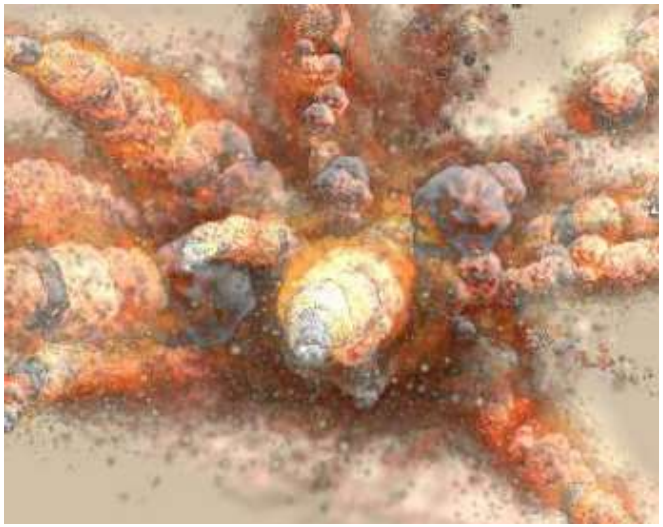
Motion Capture

- Most of the work is done by Human Actor
- Human joint angles are taken from a live model via sensors on a motion capture suit worn by the actor.
- The actor is filmed performing the scene.
- Image processing techniques are then used to identify the positions of the markers in each frame of the film, and their positions are translated to coordinates.
- These coordinates are used to determine the positioning of the body of the animated character.
- The movement of each marker from frame to frame in the film is tracked and used to control the corresponding movement of the animated character.

Procedural animation

- Most of the work is done by Computer
- Object positions and shapes are controlled by :
  - Particle, flocking and crowd systems
  - Cloth, Fire, Smoke and Water simulations





Virtual reality

Introduction

- Virtual Reality (VR) is the illusion of a three-dimensional, interactive, computer-generated reality where sight, sound, and sometimes even touch are simulated to create pictures, sounds, and objects that actually seem real.
- VR must allow the user to view the environment from any point and at any angle
- VR must allow the user to interact with objects in the environment.
- Control of and interaction with the environment is typically through direct manipulation of objects in the real world.

Types of VR System

- **Windows on World(WoW)**
  - Also called Desktop VR.
  - Using a conventional computer monitor to display the 3D virtual world.
  - Low cost, low performance and less immersive
- **Immersive VR**
  - Completely immerse the user's personal viewpoint inside the virtual 3D world.
  - The user has no visual contact with the physical world.
  - Often equipped with a Head Mounted Display (HMD).

Types of VR System

- **Telepresence**
  - A variation of visualizing complete computer generated worlds.
  - Links remote sensors in the real world with the senses of a human operator. The remote sensors might be located on a robot. Useful for performing operations in dangerous environments.
  - Real-time telepresence
    - Interactions are reflected to some real world objects.
  - Delayed telepresence
    - Interactions are recorded, and after satisfaction is applied to the real-world object.

Types of VR System

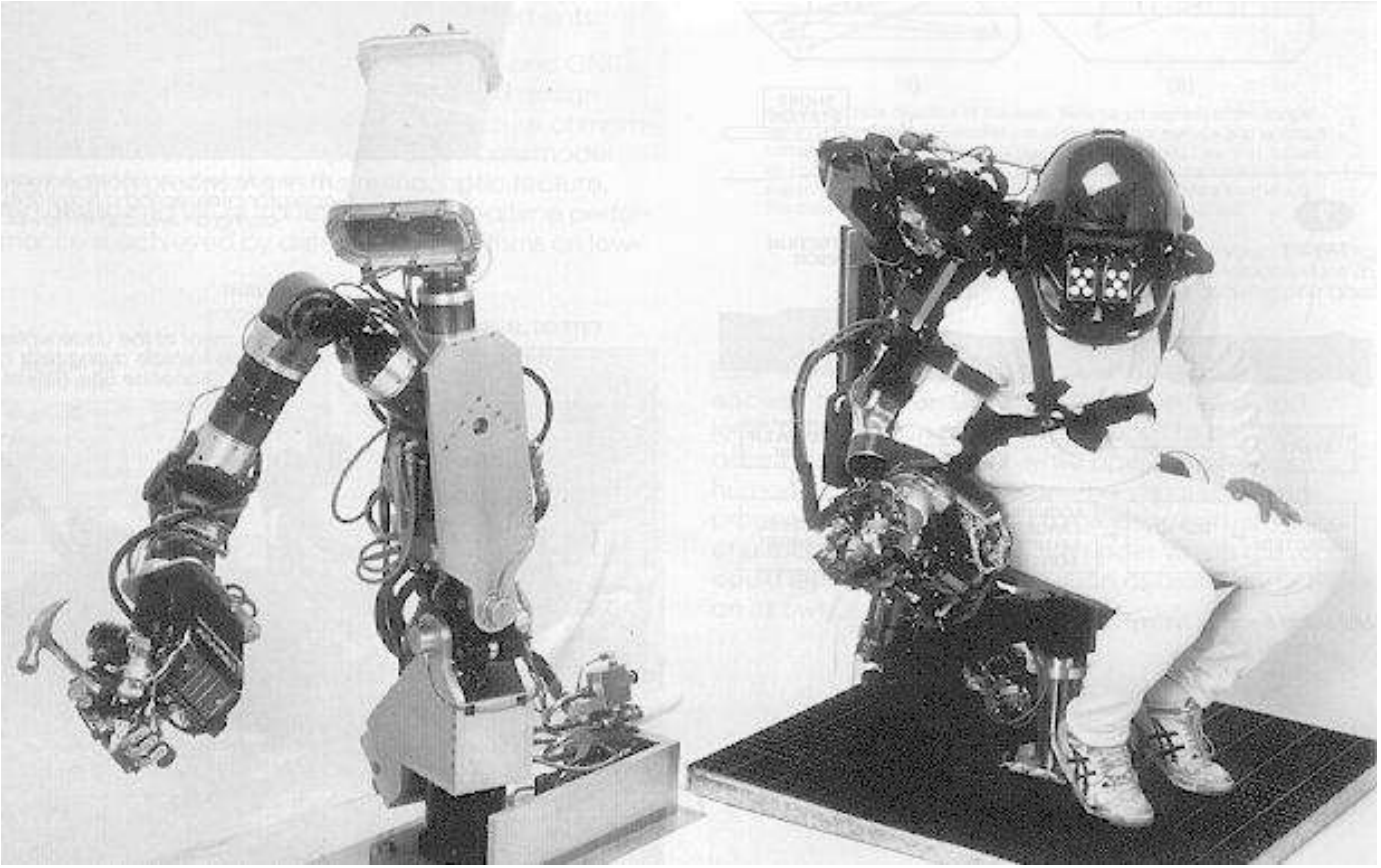
- **Mixed Reality(Augmented Reality)**
  - The seamless merging of real space and virtual space.
  - Integrate the computer-generated virtual objects into the physical world which become in a sense an equal part of our natural environment.

Distributed VR

- A simulated world runs on several computers which are connected over network and the people are able to interact in real time, sharing the same virtual world.

VR Examples

- Telepresence VR

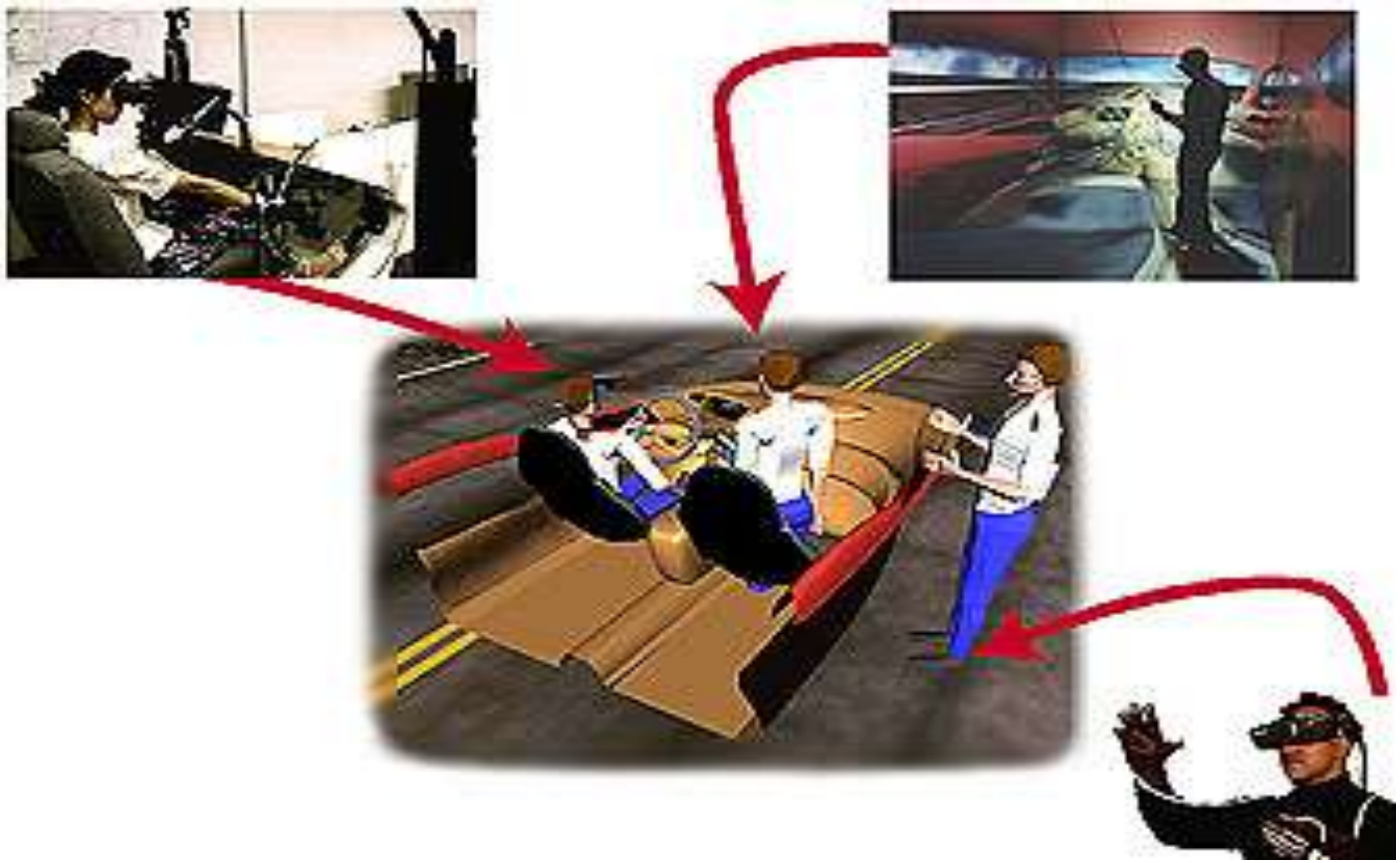


VR Examples

- Augmented VR

VR Examples

- Distributed VR



Technologies of VR--Hardware

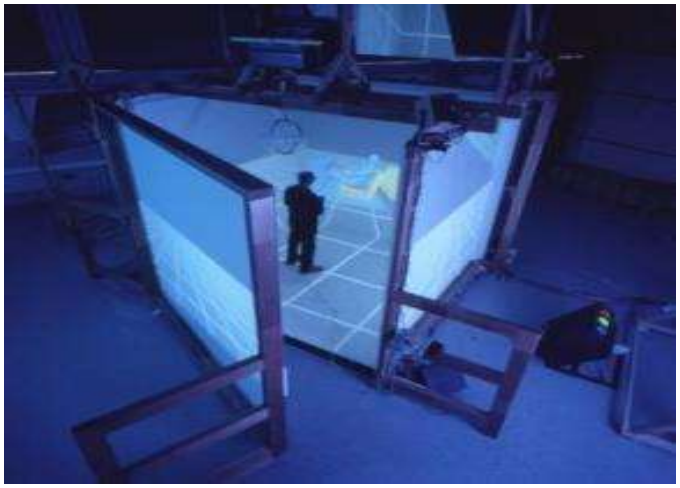
- Head-Mounted Display (HMD)
  - A Helmet or a face mask providing the visual and auditory displays.
  - Use LCD or CRT to display stereo images.
  - May include built-in head-tracker and stereo headphones

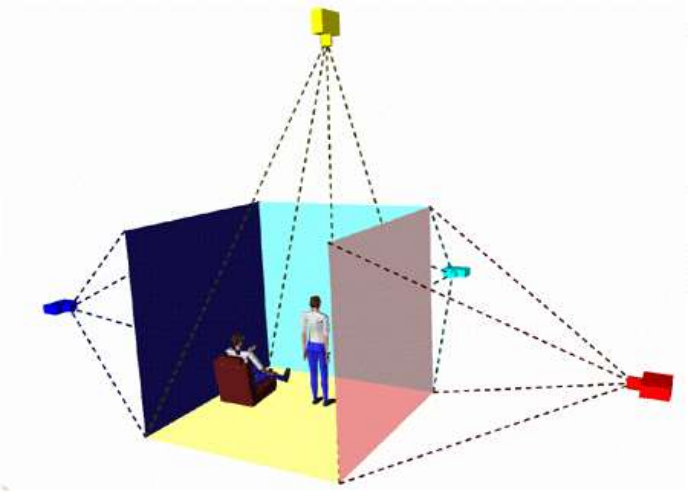
Technologies of VR--Hardware

- Binocular Omni-Orientation Monitor (BOOM)
  - Head-coupled stereoscopic display device.
  - Uses CRT to provide high-resolution display.
  - Convenient to use.
  - Fast and accurate built-in tracking.

Technologies of VR--Hardware

- **Cave Automatic Virtual Environment (CAVE)**
  - Provides the illusion of immersion by projecting stereo images on the walls and floor of a room-sized cube.
  - A head tracking system continuously adjust the stereo projection to the current position of the leading viewer.





Technologies of VR--Hardware

- **Data Glove**
  - Outfitted with sensors on the fingers as well as an overall position/orientation tracking equipment.
  - Enables natural interaction with virtual objects by hand gesture recognition.

- **Control Devices**
  - Control virtual objects in 3 dimensions.

Technologies of VR--Software

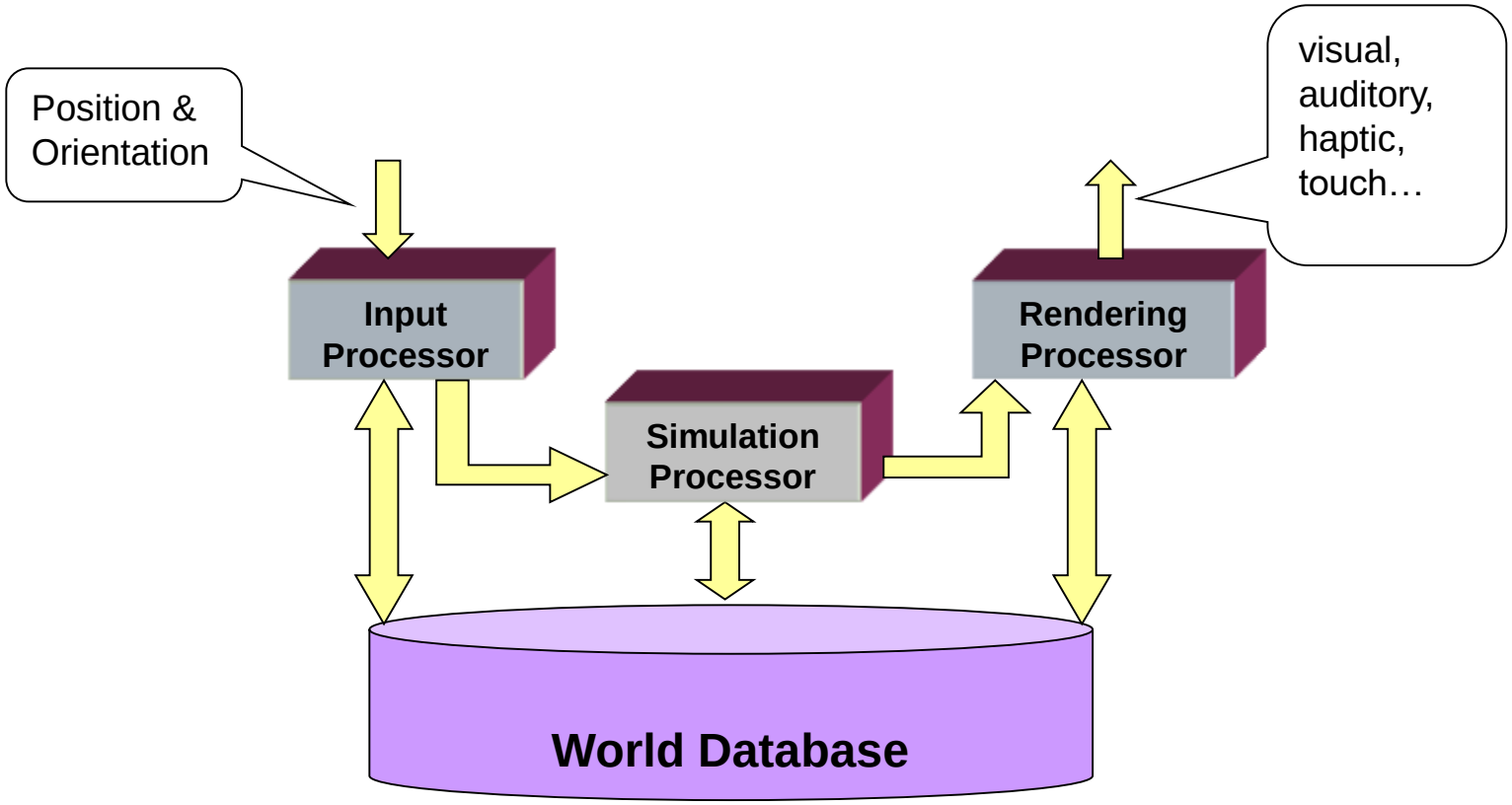
- Toolkits
  - Programming libraries.
  - Provide function libraries (C & C++).
- Authoring systems
  - Complete programs with graphical interfaces for creating worlds without resorting to detailed programming.
- Software packages available in market
  - **Multiverse (Freeware)**
  - **Virtual Reality Studio** ($100)
  - **Sense8 World Tool Kit (WTK)** (over $1000)
  - **Autodesk Cyberspace Development kit** (over $1000)

Technologies of VR--Software

- VRML(Virtual Reality Modeling Language)
  - Standard language for interactive simulation within the World Wide Web.
  - Allows to create "virtual worlds" networked via the Internet and hyperlinked with the World Wide Web.
  - Aspects of virtual world display, interaction and internetworking can be specified using VRML without being dependent on special gear like HMD.
  - VR models can be viewed by Netscape or IE with a browser plug-in.

Architecture of VR System

- Input Processor, Simulation Processor, Rendering Processor and World Database.



Components of VR System

- **Input Processor**
  - Control the devices used to input information to the computer. The object is to get the coordinate data to the rest of the system with minimal lag time.
  - Keyboard, mouse, 3D position trackers, a voice recognition system, etc.

- **Simulation Processor**
  - Core of a VR system.
  - Takes the user inputs along with any tasks programmed into the world and determine the actions that will take place in the virtual world.

Components of VR System

- **Rendering Processor**
  - Create the sensations that are output to the user.
  - Separate rendering processes are used for visual, auditory, haptic and other sensory systems. Each renderer take a description of the world stat from the simulation process or derive it directly from the World Database for each time step.

- **World Database (World Description Files)**
  - Store the objects that inhabit the world, scripts that describe actions of those objects.

Graphical User Interface (GUI)

- System users often judge a system by its interface rather than its functionality
- A poorly designed interface can cause a user to make catastrophic errors

Advantages of UIs

- They are easy to learn and use.
  - Users without experience can learn to use the system quickly.
- The user may switch quickly from one task to another and can interact with several different applications.
  - Information remains visible in its own window when attention is switched.
- Fast, full-screen interaction is possible with immediate access to anywhere on the screen

Applications of VR

- **Entertainment**
  - More vivid
  - Move exciting
  - More attractive
- **Medicine**
  - Practice performing surgery.
  - Perform surgery on a remote patient.
  - Teach new skills in a safe, controlled environment.
- **Manufacturing**
  - Easy to modify, Low cost, High efficient
- **Education & Training**
  - Driving simulators, Flight simulators, Ship simulators.